

反之, $\forall f \in (c_0)^*$, 令 $x(m, n) = f(e_{m,n})$, 于是 $\forall y = \{y(m, n)\} \in c_0$, $f(y) = \sum x(m, n)y(m, n)$. 选取 $y_N = \{y_N(m, n)\} \in c_0$ 如下:

$$y_N(m, n) = \begin{cases} e^{-i \arg x(m, n)}, & 1 \leq m, n \leq N, \\ 0, & \text{其他.} \end{cases}$$

于是

$$\sum_{n,m=1}^N |x(m, n)| = f(y_N) \leq \|f\| \|y_N\|_\infty = \|f\|.$$

因而 $\sum_{n,m=1}^\infty |x(m, n)| \leq \|f\|$. 所以 $x = \{x(m, n)\} \in l^1$, 且 $f = f_x$, $\|f_x\| = \|x\|_1$.

由此可得 $l^1 = (c_0)^*$.

(2) 设 $x_N \in M$, $x_N \rightarrow x_0$, 即当 $N \rightarrow \infty$ 时,

$$\|x_N - x_0\|_1 = \sum_{n,m=1}^\infty |x_N(m, n) - x_0(m, n)| \rightarrow 0.$$

于是

$$\begin{aligned} \lim_{N \rightarrow \infty} |x_N(m, 1) - x_0(m, 1)| &= 0, \\ \lim_{N \rightarrow \infty} \left| \sum_{n=2}^\infty x_n(m, n) - \sum_{n=2}^\infty x_0(m, n) \right| &\leq \|x_n - x_0\|_1 \rightarrow 0. \end{aligned}$$

所以

$$\lim_{N \rightarrow \infty} m x_N(m, 1) = \lim_{N \rightarrow \infty} \sum_{n=2}^\infty x_N(m, n) = \sum_{n=2}^\infty x_0(m, n) = m x_0(m, 1).$$

因此 $x_0 \in M$, M 是 l^1 的闭子集.